RAGX

Frontend & Backend Languages and Technology Stack Specification

This document provides the official technical specification of the programming languages, frameworks, libraries, databases, vector stores, build systems, and runtime environments implemented in the **RAGX** project codebase.

CALLOUT: PROJECT FREEZE VERIFICATION: Source code is 100% frozen. Zero changes were made to existing frontend or backend source files during the generation of this document.

1. Backend Programming Languages & Technology Stack

Component / Layer	Technology / Language	Details & Implementation Purpose in RAGX
Primary Language	**Python 3.11+** (.py)	Core language for backend services, REST API routing, machine learning integration, evaluation logic, and scientific matrix math.
Web Framework	**FastAPI** (>=0.100.0 / v0.110+)	Asynchronous ASGI web framework executing REST API routes with native Pydantic schema validation.
ASGI Web Server	**Uvicorn** (>=0.22.0)	High-performance ASGI server executing the FastAPI application bound to Port 8000.
Embedding Model	**SentenceTransformers** (all-MiniLM-L6-v2)	Generates 384-dimensional dense vector embeddings backed by CPU PyTorch.
Vector Database	**ChromaDB** (>=0.4.0)	In-process persistent HNSW graph vector index tagged strictly with user owner_id metadata.
Scientific Math	**NumPy**	Vectorized 2D array dot product math (sim_matrix = norm_embs . norm_embs^T) enabling ~2ms vector redundancy matrix audits.
PDF Processing	**PyMuPDF (fitz)** (>=1.22.0)	High-speed PDF parser extracting page text while preserving page number metadata.
Relational DB	**SQLite 3** (WAL Mode)	Write-Ahead Logging database managing users, active sessions, and security activity logs.
Security & Hashing	**PBKDF2 HMAC SHA-256**	Password hashing with 16-byte salt and 24-byte hex session tokens cached in RAM (<0.1ms).

2. Frontend Programming Languages & Technology Stack

Component / Layer	Technology / Language	Details & Implementation Purpose in RAGX
Primary Language	**JavaScript (ES6+) / JSX** (.jsx, .js)	Client-side programming language for component rendering, state management, and async API calls.
UI Framework	**React 19** (react^19.2.8)	Single Page Application (SPA) library utilizing functional components and custom hooks.

Build Tool & Server	**Vite 8** (vite^8.2.0)	Ultra-fast bundler and HMR dev server providing fast production compilation (2.31s).
Styling Framework	**Tailwind CSS v4** (tailwindcss^4.3.3)	Utility-first CSS styling powering dark slate/navy theme across all UI workspaces.
UI Component Icons	**Lucide React** (lucide-react^1.31.0)	Icon set used across navigation bars, evaluation badges, and administrative modals.
3D Graphics Canvas	**HTML5 WebGL 2.5D/3D Canvas**	Embedded canvas rendering engine (ThreeRAGXIntro) powering 3D introduction graphics.
HTTP API Client	**Axios** (axios^1.19.0)	Asynchronous HTTP client with authorization interceptors injecting Bearer tokens into FastAPI calls.
State Persistence	**Browser LocalStorage**	Persists active workspace tabs, user auth sessions, and evaluation reports across refreshes.

3. Summary of Codebase File Extensions

* **Backend File Types:** .py (Python source code), .db (SQLite database files), .json (Configuration & document registry data).
* **Frontend File Types:** .jsx (React JSX components), .js (JavaScript helper modules), .css (Tailwind styling), .json (npm manifest & config).
* **Repository Status:** On branch main (clean, 100% frozen, 0 uncommitted changes to source code).